

difference is that instead of the product being discarded and thrown away, it is recycled in different forms.

You need to understand the product lifecycle in detail to understand the impact it has on the environment when it is used. So look at a slightly detailed product lifecycle in Fig. 3. It shows the different phases that the product goes through in its entire life. The flow diagram has seven processes. While six of them are sequential, the seventh is a concurrent one. Let us see what each of these phases means.

Concept generation. Typically, this phase involves creating concept based on the inputs given. It is predominantly more creative and needs trained industrial designers to successfully create product concepts.

Design. This is the phase where the concepts are physically realised. Depending on the complexity of the product, either the actual implementation or proof of concept is created. In the case of proof of concept, typically, the product undergoes more rounds of user validation to check the product features before the actual design is done. Reason for this approach is that when the product is complex and

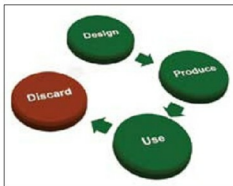


Fig. 1: Conventional product lifecycle where the product is discarded after use



Fig. 2: Future product lifecycle where product is recycled after use

needs custom tooling, it is wiser to verify the concept before making any serious investment into the product.

Engineering phase. For products that don't need proof of concept, design and engineering phases are integrated. In

As there is little control once the product leaves the manufacturer, the raw materials shouldn't require any special process for recycling or reuse

this phase, typically, the design is implemented to meet the manufacturing, testing and supporting requirements. Engineering phase is very crucial for the product's success from the market perspective. Bad engineering can mar the product's success in the field.

New product introduction. In this phase, the product is transferred to the manufacturing line. Here the design is made to comply with all the design-for-efficient-manufacturing guidelines of the contract manufacturer, testers are built if needed, the manufacturing line team is trained and the product quality is maintained within the designated norms.

Manufacturing. This phase is typically executed by the OEM or the EMS vendors. Most product vendors (OEMs) have started outsourcing the manufacturing to contract manufacturers to get the advantages of lower cost as well as global reach.

Support. This activity is typically after-sales support for the product, which could range from customer support to supporting the product in the manufacturing line. It is an ongoing activity and stops only when the product is withdrawn from the market.

Sustain. This is a very important activity in the product lifecycle and is predominantly design-centric. While most of you will be familiar with software sustenance (as most of the revenue for India comes from this service), product sustenance encompasses all the sections of a product covering hardware, mechanical and software. Some of the critical activities that form part of sustenance are tackling component obsolescence, cost reduction exercises (which are a must as the competitors start nibbling the market

RoHS Demystified

RoHS stands for 'restriction/reduction of hazardous substances.' It is a European legislation that bans six hazardous substances from manufacturing processes: cadmium (Cd), mercury (Hg), hexavalent chromium (Cr (VI)), polybrominated biphenyls, polybrominated diphenyl ethers and lead (Pb). RoHS is also known as lead-free technology.

This legislation is effective July 1, 2006, and from this date the products using these substances cannot be sold in Europe anymore. The problem in shifting to the RoHS process is that solders traditionally use 60 per cent of tin (Sn) and 40 per cent of lead (Pb). Alternative materials have higher melting temperature: traditional tin/lead solder melts at 180°C, while lead-free solder melts at 227°C. This means that the electronic components must be able to support this new soldering temperature in order to allow lead-free solder to be used and also the PCB layout has to be designed to support higher viscosity of the lead-free solders. The second biggest problem is the tin whisker growth in the lead-free solders which can create potential shorts in the field after some time.

The shift in process is basically with the electronic waste. A lot of electronic equipment are ending their lives in open junkyards and waste dumps all around the world—many of them with no chemicals control. The water from acid rain dissolves lead and other hazardous materials from the electronic equipment, and the rainwater mixed with these materials goes straight to the water table and then to the drinking water and pollutes that.

Lead can affect almost every organ and system in the body, especially the central nervous system. Kidneys and the reproductive system are also affected. The effects are the same, whether lead is breathed or swallowed. At high levels, lead may decrease reaction time, cause weakness in fingers, wrists or ankles, and possibly affect the memory. It may also cause anaemia.